

Easy IP Setup Tool — User Guide

A cross-platform Python TUI (Terminal User Interface) for discovering and managing i-PRO IP cameras and recorders on your network.

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Getting Started

Requirements

- Python 3.8 or higher
- Required packages (install via `pip install -r requirements.txt`):
 - textual
 - rich

Running the Application

```
python easy_ip_tui.py
```

Or on Windows, double-click `easy_ip.bat`.

Main Interface

When you launch the application you will see a menu bar at the top, the device table in the centre, and a status bar at the bottom.

Menu Bar

Button	Key	Purpose
File	F1	Load / save site configurations
Scan	F2	Discover devices on the network
Monitor	F3	Start / stop continuous monitoring
Groups	F4	Manage device groups
Setup	F5	Configure application settings
Config IP	I	Change IP addresses for tracked devices
Admin PW	A	Set administrator credentials on unconfigured cameras

Status Bar

- Monitoring status (ON / OFF)
- Device counts — total, online, offline
- Last scan timestamp
- Next scan timestamp (when monitoring is active)

Theme

The application supports multiple colour themes. Open the command palette with `Ctrl+P`, type **theme**, and choose from the available options. Your chosen theme is saved automatically and restored on the next launch.

Scanning for Devices

Automatic Network Scan

- Click **Scan** or press F2
- Select **Scan Network (Auto)**
- Adjust the timeout if needed (default: 3 seconds)
- Wait for the scan to complete

The scan sends a UDP broadcast on port 10670 to discover all i-PRO cameras and recorders on your local network segment. Each camera's response also carries its current setup-window state and admin-password status, which are stored alongside the device record.

First-Time Scanning

If you scan before creating any groups, the application automatically creates a **"Default"** group so discovered devices have somewhere to go immediately.

Scan Results

After scanning you will see:

- Number of devices found
- New devices not yet tracked
- Devices already in your site

Select which new devices to add and choose a target group.

After adding devices, if any camera has no admin password set, a warning notification is shown listing those cameras by name. This is a reminder to set credentials before putting the cameras into service.

Manual Device Addition

If a device is not discovered automatically:

- Click **Scan > Manual Add Device...**
- Enter the device details:
 - Device Name
 - MAC Address (format: aa : bb : cc : dd : ee : ff)
 - IP Address
 - Model (optional)
 - Select a target group
- Click **Add**

Device Security Status

Each camera reports two security-related states in every discovery response. The application tracks and displays these automatically.

Setup Window

i-PRO cameras accept IP configuration changes only within a **20-minute window** after power-on or factory reset. Outside this window the camera silently ignores all configuration commands.

State	Meaning
Setup window open	Camera will accept IP configuration changes
Setup window expired	Camera will reject IP changes — power-cycle to reopen

The current state is shown in the **Device Details** dialog (press Enter on any device row).

When you attempt to configure a camera whose setup window has already expired, the application detects this during its pre-scan and reports the failure immediately — no packets are sent to the camera.

Admin Password

i-PRO cameras ship without an administrator account. Until a password is set, the camera's web interface and certain management functions are unsecured.

Indicator	Meaning
[No PW] in device name (yellow)	No admin password is set
"Admin Password: NOT SET" in Device Details	Same — visible when you press Enter on that row
Orange warning after a scan	Lists all tracked cameras currently without a password

Both states refresh automatically on every scan — once a password is set, the warning clears at the next scan without any manual action.

Configuring IP Addresses

The Configure IP dialog lets you change the network settings of one or more tracked cameras in a single operation. Press **I** or click **Config IP** to open it.

Important: Before sending any configuration the application performs a quick network scan (approx. 2 seconds) to check each camera's current setup-window state. Cameras outside their setup window are skipped and reported as failed — they will not waste time sending packets that the camera would ignore.

Network Mode

Select the addressing mode from the dropdown at the top of the dialog:

Mode	Description
Static IP	The camera uses the IP address you specify permanently.
DHCP	The camera requests an IP address from a DHCP server.
Auto (AutoIP)	DHCP with automatic link-local fallback (169.254.x.x) if no DHCP server responds.
Auto (Advanced)	DHCP with enhanced auto-negotiation and AutoIP fallback.

Shared Network Settings

These values apply to every camera you configure in this session:

- **Subnet Mask** — pre-filled from the first tracked device
- **Gateway** — pre-filled from the first tracked device

DNS Settings

Choose how each camera resolves hostnames:

- **Auto** — the camera obtains DNS addresses from DHCP or the network automatically.
- **Manual** — enter a Primary DNS and Secondary DNS address explicitly.

Assigning IP Addresses (Static mode)

Auto-assign from a base address:

- Enter a starting IP (e.g. 192.168.1.100) in the **Auto-assign** field
- Click **Auto-assign** 
- Each ticked camera receives the next sequential address (100, 101, 102 ...)
- The base address field advances automatically to the next available address
- The next time you open the dialog the field is pre-filled with that address, ready for the next batch

Manual entry:

Type directly into the **New IP** field beside each camera row.

Select All / Select None:

Use these buttons to tick or clear all cameras at once.

Applying Settings

Click **Apply Selected** to send the new configuration to every ticked camera. A progress bar shows each camera being configured. The application uses the same two-phase broadcast protocol as the original i-PRO EasyIP.exe tool — the config packet is sent three times followed by a commit packet, exactly mirroring the original behaviour.

The progress screen reports results in three categories:

Result	Meaning
Succeeded	Camera accepted the new configuration
Setup window expired	Camera was outside its setup window — power-cycle and retry
Failed	A network or protocol error occurred

Setting Administrator Credentials

i-PRO cameras ship with no administrator account. The **Admin PW** function lets you register credentials on multiple cameras in a single operation, using the camera's built-in `/cgi-bin/reg_admin` CGI endpoint.

Note: This function only works on cameras that have not yet had an admin password set. Once a password is registered, changing or resetting it must be done through the camera's web interface.

Opening the Dialog

Press A or click **Admin PW** in the menu bar. The application automatically filters the device list to show only cameras where [No PW] is active. If all tracked cameras already have passwords, a notification says so and the dialog does not open.

Dialog Layout

[illegible]

All cameras without a password are pre-ticked. Use **All** / **None** to adjust the selection.

Password Requirements

- 8 to 32 characters

- Must contain at least one letter (a–z or A–Z)
- Must contain at least one digit (0–9)

The application validates these rules before making any network calls.

What Happens on Apply

For each selected camera the application:

- Connects over HTTPS to the camera's web port (falls back to HTTP if needed)
- Sends the credentials as a POST request with BASE64-encoded parameters
- Reports success or the specific error for each camera individually

On success the [No PW] marker in the device table disappears immediately — no rescan needed.

Possible Outcomes

Response	Meaning
Completion of registration	Credentials set successfully
Administrator already registered	A password was already set — use the web interface to change it
Invalid value	Password does not meet the requirements
Connection failed	Camera unreachable — check IP address and network path

Managing Groups

Groups organise your devices by location, floor, function, or any criteria you choose.

Creating a Group

- Click **Groups** or press F4
- Select **Add New Group...**
- Enter a group name
- Click **Create**

Removing a Group

- Click **Groups > Remove Group...**
- Select the group to remove
- Click **Remove**

Warning: Removing a group permanently deletes all devices in that group from the site.

Moving Devices Between Groups

The Move Devices dialog supports moving multiple cameras at once — essential when reorganising a large site.

- Click **Groups > Move Device...**
- The dialog lists every tracked device with its current group shown in parentheses
- Tick the devices you want to move — use the helper buttons for speed:

Button	Effect
All	Selects every device in the list
None	Clears all selections
Invert	Flips every checkbox — useful when most cameras need to move and only a few should stay

- Choose the target group from the dropdown
- Click **Move Selected**

A single notification confirms how many devices were moved and to which group.

Group Status Colours

Colour	Meaning
Green	All devices online
Red	All devices offline
Orange	Mix of online and offline
Grey	Group is empty

Expanding / Collapsing Groups

- Click a group header to expand or collapse it
- Press Space when a group row is focused
- Expanded groups show [-], collapsed groups show [+]

Monitoring Devices

Monitoring periodically scans the network and updates each device's online / offline status automatically.

Starting Monitoring

- Click **Monitor** or press F3
- The menu bar label changes to **MONITORING: ACTIVE** (green)
- An immediate scan runs
- Subsequent scans run at the interval set in Setup

Status Bar When Monitoring

[MONITORING: ON] | Devices: 7 (Online: 6, Offline: 1) | Last: 14:30:15 | Next: 14:31:15

Stopping Monitoring

Click **Monitor** or press F3 again.

Offline Alerts

When a monitoring scan detects offline devices you receive a notification:

Monitor scan: 2 device(s) offline

Deep Check

When **Deep Check** is enabled in Setup, each scan also queries the HTTP webserver on every camera to confirm it is genuinely responding — not just present on the network. This catches cameras that are reachable via ping but have a crashed webserver.

Setup & Configuration

Access settings via **Setup** or press F5.

Site Name

Give your site a descriptive name such as "Main Office" or "Warehouse – Level 2".

Network Interface

Important on multi-NIC systems.

If your PC has multiple adapters (Ethernet, Wi-Fi, VPN), select the one connected to the camera network:

- **All Interfaces (0.0.0.0)** — broadcast on all adapters (default)
- **Specific interface** — select the IP of the adapter on the camera subnet

Tip: If scanning finds no devices, switch from All Interfaces to your specific LAN adapter.

Deep Check

Queries the webserver on each camera after every scan to confirm it is responding. Adds a small amount of time to each scan cycle.

Scan Frequency

How often monitoring scans run, in seconds:

Network size	Recommended interval
Small (< 20 cameras)	30 – 60 s
Medium (20 – 100)	60 – 120 s
Large (100+)	120 – 300 s

Data Folder

Where site .json files are saved. Can be an absolute path or a path relative to the application directory.

Display Columns

Toggle which columns appear in the device table:

Column	Description
Device Type	Camera or Recorder
IP Address	Current IP address

Column	Description
MAC Address	Hardware (MAC) address
Model	Device model number
Serial Number	Device serial number
Status	Online / Offline / Unknown
Firmware	Firmware version string
HTTP Port	Web interface port number

File Operations

Saving Your Site

- Click **File** or press F1
- Select **Save Site** (or **Save Site As...** for a new file)
- Enter a filename
- Click **Save**

Site files are saved as `.json` and include all groups, devices, admin-password state, column visibility settings, scan frequency, and the network interface selection.

Loading a Site

- Click **File > Load Site...**
- Navigate to your site file
- Select the `.json` file
- Click **Load**

Tip: The application does not auto-save. Save before exiting to retain any changes.

Exporting Data

Export device information for reports, spreadsheets, or integration with other tools. Press E or use the File menu.

Export Format

Format	Best for
JSON	Scripting, APIs, structured data
CSV	Excel, Google Sheets, reporting tools

Export Scope

- **All Groups** — every tracked device
- **Selected Group** — one group only

Exportable Fields

Device Name, Device Type, IP Address, Subnet Mask, Gateway, MAC Address, Model, Serial Number, HTTP Port, Firmware Version, Network Mode, Status, First Seen, Last Seen.

Keyboard Shortcuts

Key	Action
F1	File menu
F2	Scan menu
F3	Toggle monitoring
F4	Groups menu
F5	Setup menu
I	Configure IP addresses
A	Set administrator credentials (Admin PW)
E	Export devices
R	Refresh display
O	Open selected device in browser
Space	Expand / collapse group
Enter	View device details
C	Copy selected cell value
Ctrl+P	Command palette (theme, commands)
Q	Quit application
Esc	Close current dialog

Command Line Usage

The underlying engine can also be used without the TUI.

Discover Devices

```
python Easy_IP.py discover --table
python Easy_IP.py discover --interface 192.168.1.10 --table
python Easy_IP.py discover --json > devices.json
python Easy_IP.py discover --csv > devices.csv
python Easy_IP.py discover --table --sort type
```

Sort options: ip (default), mac, serial, type

Discovery output now includes setup window state and admin password status for each device.

Configure a Device

```
# Static IP with manual DNS
python Easy_IP.py configure \
--mac d4:2d:c5:2a:8d:13 \
--ip 192.168.1.54 \
--subnet 255.255.255.0 \
--gateway 192.168.1.1 \
--mode static \
--dns-mode manual \
--primary-dns 192.168.1.1 \
--secondary-dns 8.8.4.4

# DHCP with auto DNS
python Easy_IP.py configure \
--mac d4:2d:c5:2a:8d:13 \
--ip 192.168.1.54 \
--subnet 255.255.255.0 \
--gateway 192.168.1.1 \
--mode dhcp \
--dns-mode auto

# Auto (AutoIP) mode
python Easy_IP.py configure ... --mode auto_autoip

# Auto (Advanced) mode
python Easy_IP.py configure ... --mode auto_advanced
```

The configure command auto-discovers the camera first. If the camera's setup window has expired the command exits immediately with an error rather than sending packets the camera would ignore.

Network Mode Options

--mode value	Description
static	Static IP (default)
dhcp	DHCP
auto_autoip	Auto with AutoIP fallback
auto_advanced	Auto Advanced

DNS Mode Options

--dns-mode value	Description
manual	Use --primary-dns and --secondary-dns (default)
auto	Camera obtains DNS automatically

Set Administrator Credentials

```
python Easy_IP.py set-admin \
--ip 192.168.1.101 \
--port 443 \
--username admin \
--password Admin1234
```

Only succeeds when the camera has no admin registered. Password must be 8-32 characters and contain both letters and numbers.

Diagnostics

```
python Easy_IP.py diag
python Easy_IP.py discover -v # verbose – shows raw protocol detail
```

Troubleshooting

No Devices Found

Check the network interface

Go to Setup > Network Interface. Select your specific LAN adapter instead of All Interfaces. Cameras must be on the same subnet as the selected adapter.

Check the firewall

Allow Python through Windows Firewall. UDP ports 10669 and 10670 must be open for broadcast traffic.

Increase the scan timeout

In the Scan menu, increase the timeout to 5 – 10 seconds. Large or slow networks may need more time.

Verify basic connectivity

Confirm you can ping a camera's IP address from the PC running the tool.

Devices Show as Offline

- Verify the device is powered on and the network cable is connected
- Check that the IP address has not changed (use a DHCP reservation for stable IPs)
- Run a manual scan to refresh status immediately

IP Configuration Has No Effect

Setup window may have expired

i-PRO cameras only accept IP changes within 20 minutes of being powered on. If the setup window has expired, the device table will show this in the Device Details dialog. Power-cycle the camera and retry within 20 minutes.

Port mismatch

The camera's HTTP port in the config packet must match its current port. The tool auto-detects this via discovery. If auto-detection fails, specify `--port` explicitly on the command line.

Broadcast blocked

Ensure UDP broadcast is not blocked between the PC and the camera subnet. Some managed switches block broadcast traffic between VLANs.

Cannot Set Administrator Password

Camera already has a password

The `reg_admin` CGI endpoint returns HTTP 503 if an admin is already registered. Use the camera's web interface (`https://<camera-ip>`) to manage credentials on cameras that already have a password.

Connection fails

- Confirm the camera's IP address is correct and reachable (ping test)
- The tool tries HTTPS on the configured port then HTTP on port 80 automatically
- Check that TCP port 443 (or 80) is not blocked by a firewall between the PC and camera

Password rejected

Ensure the password is 8-32 characters and contains at least one letter and at least one digit. Spaces and some special characters may also cause issues.

Application Won't Start

Verify Python 3.8+ is installed:

```
python --version
```

Install required packages:

```
pip install -r requirements.txt
```

Interface Selection Shows Wrong Adapters

- Ensure the adapter is enabled and has an IP address assigned
- Restart the application after connecting to a new network
- If psutil is installed, interface detection is more accurate:

```
pip install psutil
```

Support

For issues and feature requests, visit:

https://github.com/Rickardjd/Easy_IP

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